Javadoc Pattern Cards

Task 2 · Source: javadoc_events CSV (events cl-0001 to cl-0028)

Overview

Each card describes one pattern, not one event, so events that show the same pattern share a card. Cards marked **(candidate)** are based only on release-note text in the CSV; the actual Javadoc diff still needs to be checked before the card is treated as a confirmed rule. The "smallest trigger" lines are written as concrete, checkable conditions so each can become an automatic mutation rule.

Pattern cards

Card 1: Summary describes the wrong behavior

- **Event ID(s):** cl-0002
- **Short name:** Summary describes the wrong behavior
- **What happened in code:** The method's actual test changed or was clarified: it checks that the string contains only '0' characters.
- **What happened in doc:** The summary said "Returns true if s is null or empty."
- **Why the doc is wrong:** The caller expects a null/empty check but gets a different predicate.
- **Smallest trigger:** The method's condition or return logic changes while the first-sentence summary still describes the old condition.

Card 2: Circular or vague summary

- **Event ID(s):** cl-0001
- **Short name:** Circular or vague summary
- **What happened in code:** No behavior change; the method gets a clearer description.
- **What happened in doc:** The summary was "Utility method for {@link #createNumber(String)}", which points at itself and says nothing concrete.
- **Why the doc is wrong:** The reader learns nothing about what the method returns.
- **Smallest trigger:** Replace the summary sentence with a self-referencing or generic "Utility method for ..." sentence.

Card 3: Malformed inline HTML tag

- **Event ID(s):** cl-0005
- **Short name:** Malformed inline HTML tag
- **What happened in code:** None.
- **What happened in doc:** An opening tag was closed with </am>.
- **Why the doc is wrong:** Javadoc renders badly or fails doclint.
- **Smallest trigger:** Any inline HTML tag whose closing tag is misspelled, wrong, or missing.

Card 4: Javadoc link changed so it may no longer resolve

- **Event ID(s):** cl-0003, cl-0008
- **Short name:** Javadoc link changed so it may no longer resolve
- **What happened in code:** Class or package context (imports, location) changed or was assumed.
- **What happened in doc:** {@link org.apache.commons.lang3.builder.HashCodeBuilder} became {@link lang3.builder.HashCodeBuilder}.
- **Why the doc is wrong:** The link target resolves differently or not at all.
- **Smallest trigger:** Truncate or alter the package path in a {@link}/{@see} target so it no longer matches an import or a real class.

Note: cl-0008 (DurationFormatUtils) has the same old/new text as cl-0003 (HashCodeBuilder test). It looks like a copy-paste error, so check the source.

Card 5: Wrong version or number literal in documentation

- **Event ID(s):** cl-0006
- **Short name:** Wrong version/number literal in documentation
- **What happened in code:** The dependency was bumped to commons-parent 98.
- **What happened in doc:** changes.xml recorded 97.
- **Why the doc is wrong:** The documented value disagrees with the real one.
- **Smallest trigger:** A literal (version, count, constant) in docs differs from the value in code or build after a change.

Card 6: Javadoc missing entirely on API elements

- **Event ID(s):** cl-0013

- **Short name:** Javadoc missing entirely on API elements
- **What happened in code:** New or Java 21-related public APIs exist.
- **What happened in doc:** There were no Javadoc blocks.
- **Why the doc is wrong:** The public API is undocumented.
- **Smallest trigger:** Delete the whole Javadoc block of a public method or class.

Card 7: Documented behavior doesn't match actual behavior

- **Event ID(s):** cl-0018
- **Short name:** Documented behavior doesn't match actual behavior
- **What happened in code:** Parser behavior differs in affected releases (issue #1033).
- **What happened in doc:** The class doc still says it accepts comma elision and inserts null.
- **Why the doc is wrong:** The caller relies on a documented leniency that isn't there.
- **Smallest trigger:** Change or remove a code branch that a doc sentence describes; leave the sentence unchanged.

Card 8: Missing caveat or limitation note

- **Event ID(s):** cl-0019, cl-0020
- **Short name:** Missing caveat/limitation note
- **What happened in code:** No change; Precision.round has an inherent double-representation limit.
- **What happened in doc:** The limitation wasn't described (NUMBERS-199) and was added later.
- **Why the doc is wrong:** The caller assumes exact decimal rounding.
- **Smallest trigger:** Delete a caveat or @implNote sentence that warns about precision or edge-case limits.

Note: cl-0020 duplicates cl-0019 word for word, so the two are merged.

Card 9 (candidate): Edge-case semantics changed, doc keeps old semantics

- **Event ID(s):** cl-0021, cl-0022, cl-0023 (also cl-0011 as supporting evidence)
- **Short name:** Edge-case semantics changed, doc keeps old semantics

- **What happened in code:** compareTo was fixed for negative fractions (NUMBERS-207) and for NaN handling (Precision.compareTo); cl-0011 shows a test expecting NaN changed to positive infinity.
- **What happened in doc:** Unknown. The doc may still describe the old ordering or result.
- **Why the doc is wrong:** The documented ordering or special-value result no longer holds.
- **Smallest trigger:** A comparison or parse special case (negative, NaN, infinity) changes in code; the sentence describing that case is untouched.

Card 10 (candidate): Stated input range or bound is stale

- **Event ID(s):** cl-0024, cl-0026
- **Short name:** Stated input range/bound is stale
- **What happened in code:** A minimum epsilon bound was added (cl-0024); the supported input range was widened to the full [0, 2] (cl-0026).
- **What happened in doc:** The old range or constraint text remains.
- **Why the doc is wrong:** The caller is told the wrong valid inputs.
- **Smallest trigger:** A guard, constant, or bound in code changes; the range stated in @param or the description keeps the old value.

Card 11 (candidate): Deprecation not reflected in Javadoc

- **Event ID(s):** cl-0025
- **Short name:** Deprecation not reflected in Javadoc
- **What happened in code:** The class was deprecated and caching removed (NUMBERS-178).
- **What happened in doc:** It was presented as a normal supported class.
- **Why the doc is wrong:** The caller isn't warned and doesn't know the replacement.
- **Smallest trigger:** An @Deprecated annotation is added without a matching @deprecated tag (or the tag is removed).

Card 12 (candidate, weak): Implementation changed, descriptive doc may now be inaccurate

- **Event ID(s):** cl-0027, cl-0028
- **Short name:** Implementation changed, descriptive doc may now be inaccurate

- **What happened in code:** BrentSolver got an overflow-safe midpoint; GCD and unsigned operations were optimized.
- **What happened in doc:** Unknown. This only matters if the doc describes the old algorithm or limits.
- **Why the doc is wrong:** The doc describes a mechanism or limit that no longer applies.
- **Smallest trigger:** An algorithm-level change under a method whose Javadoc describes how it computes the result.

This is the lowest-priority card. Keep it only if the diffs show Javadoc that describes the algorithm.

Style-only changes (benign controls, not faults)

- **cl-0004, cl-0007:** `<strong>null</strong>` replaced with `{@code null}`. If you want a style rule, mutate in the reverse direction.
- **cl-0009, cl-0010:** a formatting merge and "Checks whether" changed to "Tests whether". The old and new text in cl-0009 are identical, so check that row.
- Use these as negative examples so the tool doesn't flag harmless edits.

Not usable for Javadoc rules (project-level docs)

- **cl-0014, cl-0015:** missing external Javadoc badge. These are duplicates.
- **cl-0016, cl-0017:** release docs and release history entries.
- **cl-0012:** a mixed docs and warning-fix commit. Inspect the diff first.
- These share a pattern of "project-level doc not updated for a new release or feature". Keep that as a separate card only if the tool will also mutate README or changelog files.

Gaps

The example in the card template, "missing @throws after a new exception," does not appear in the data. cl-0007 touches @throws, but only for formatting. A new event is needed if you want that rule from real evidence.